Analysis of equipment and machines for implementation in production lines

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Abstract

Equipment and machines are essential resources in production lines. Evaluating them for implementation is therefore an important and complex responsibility for manufacturing organisations. Although each company has its own requirements and procedures, the common objective is to select equipment that delivers the expected results. Drawing on bibliographic research and an industrial example involving the inspection of electronic boards, this article examines how a structured sequence of steps and the use of analytical methods can support equipment selection. It considers process characteristics, problem identification, performance requirements, practical testing and comparative evaluation.

Keywords: Equipment selection; process analysis; evaluation methods.

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About this edition

This edition revises the English and presentation of the original article. The figures have been translated and redrawn, and the comparison table has been rebuilt for readability. The original author credits, references and anonymised company designation are retained. Translated quotations follow the English wording supplied in the original document. The illustrative matrix values are unchanged; numerical presentation is explained alongside the table.

1 Introduction

Since the Industrial Revolution, manufacturers have sought to improve productivity and operating results. Automation using technology and machinery supports consistent production, product quality and the efficient use of time and raw materials. Selecting suitable equipment is consequently an important part of manufacturing performance.

The market offers a wide range of machines and suppliers. Faced with this choice, a manufacturer must determine which equipment best meets the needs and objectives of its production line. Each alternative may perform well in different areas, making it essential to define the expected outcomes and the steps required to evaluate them.

This article examines how a sequence of evaluation stages, supported by appropriate methods and tools, can help a company make an informed equipment-selection decision. It draws on the work of Vargas, Mello, and Almeida and Neto, together with a study at a company identified as “X”. The discussion covers four areas: identifying process characteristics, visualising process problems, defining expected results, and conducting tests and comparisons.

2 Process characteristics

As markets evolve, companies increasingly face decisions involving multiple factors and constraints. Project management provides a structure for this work, although each project may have different characteristics and objectives.

> *"A project is a non-repetitive undertaking, characterized by a clear and logical sequence of events, with a beginning, middle and end, which is intended to achieve a clear and defined objective, being conducted by people within predefined parameters of time, costs, resources involved and quality." (Vargas, 2018, p. 14)*

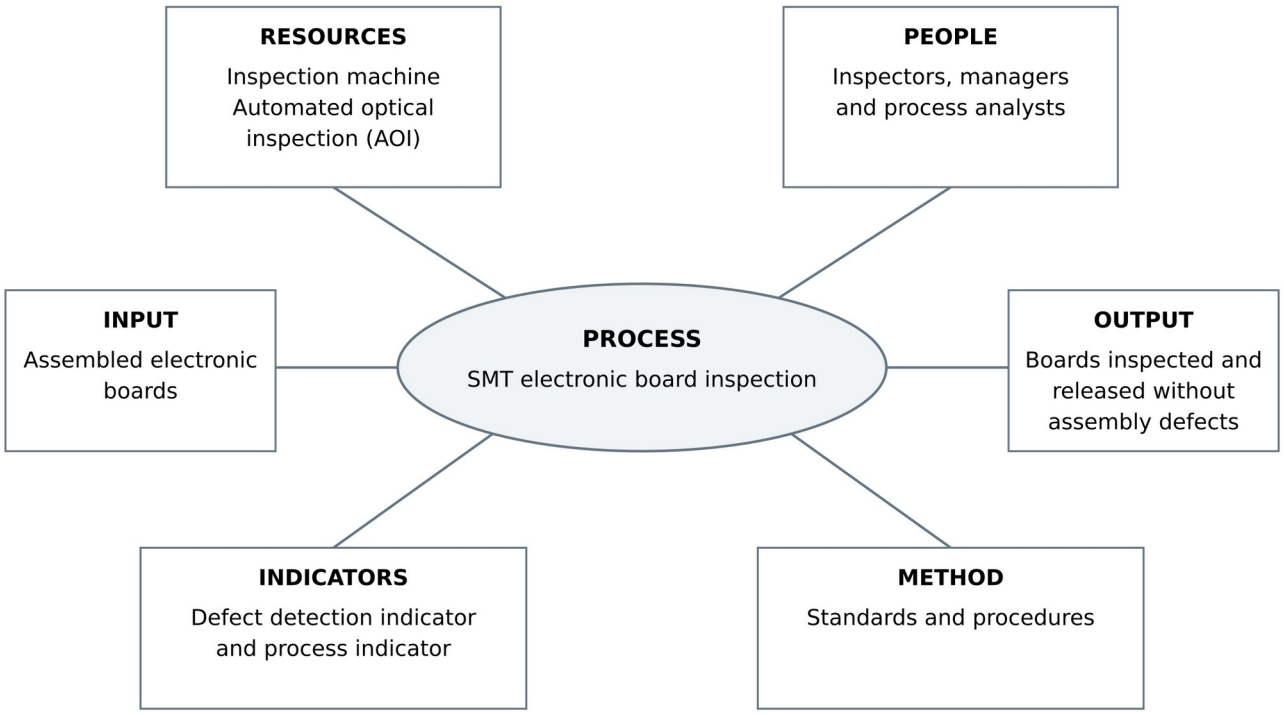
Understanding the process is the starting point. Unclear product specifications or process requirements can make a project difficult to develop and lead to unsuitable decisions or poor performance. Vargas identifies several possible causes of project failure:

> *"the goals and objectives are poorly established or are not understood by the lower echelons. The project is based on insufficient or inadequate data. The needs for personnel, equipment and materials were not known" (Vargas, 2018, p. 18).*

A turtle diagram can summarise the resources, people, methods, inputs, outputs and indicators involved in a process. It provides a practical basis for identifying what the factory needs from the equipment under evaluation. In discussing NBR 9001, Mello describes the process approach and the requirements relating to documentation, resources, product development, measurement, analysis and improvement (2011, p. 171).

Company X, located in Santa Catarina, Brazil, assembles electronic boards. The equipment evaluation concerns the inspection of boards with surface-mount technology (SMT) components. Figure 1 represents this process.

Figure 1 Turtle diagram for SMT electronic board inspection



Source: the author. English reconstruction of the original diagram.

The diagram brings together the information needed to understand the process as a whole. The principal characteristics of the production environment at Company X are:

- Around 700 different products.
- Around 30 setup changes per day.
- A shortest assembly-line cycle time of 20 seconds.
- A variety of board layouts and component types.

These figures describe the operating context; they are not measurements of improvement following equipment implementation.

2.1 Identifying process problems

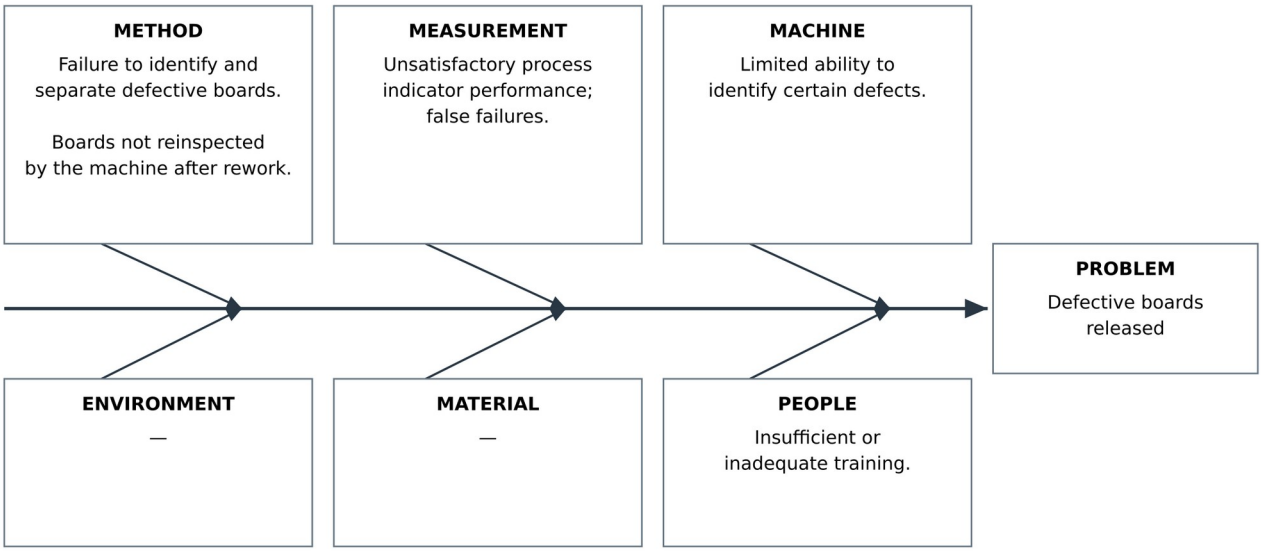
Identifying process problems helps define which requirements must be assessed and how a proposed machine could address them. An Ishikawa, or cause-and-effect, diagram provides a structured way to connect process knowledge with areas requiring investigation *and improvement.*

> *“It is used to demonstrate the relationship between the causes and effects of a process. The great advantage of the diagram is that it allows for the unfolding and*

branching of causes until reaching the one that is effectively the origin of the problem." (Mello, 2011, p. 88)

Figure 2 presents the issues identified in the inspection process at Company X. These provide a basis for evaluating equipment requirements and organising further investigation.

Figure 2 Ishikawa diagram for the electronic board inspection process



Source: the author. English reconstruction of the original diagram.

The dashes in the environment and material categories are retained from the original figure; no specific causes were entered in those categories.

One issue identified in the diagram is the inspection equipment's limited ability to detect certain defects. Relating this issue to the product and process characteristics helps distinguish mandatory requirements from desirable features. Understanding how each candidate machine works is also necessary.

Under the machine category, the evaluation must establish whether the equipment can inspect the component types and board layouts used on the production line. Equipment performance also affects the measurement category, through process indicators, and the people category, through ease of operation and the training needed for operators to adapt.

This analysis helps identify machines that could address existing problems. It can also support discussions with suppliers about modifications or improvements. Understanding the process problems provides a basis for assessing the importance and score of each evaluation criterion.

2.2 Expected results

Product quality is a continuing concern in manufacturing. The ability of equipment to meet process and customer requirements is one of several factors affecting quality performance. Vargas defines quality control as follows:

> *“Quality control: is the process of monitoring and recording the results of the execution of quality management activities to evaluate performance and ensure that project outputs are complete, correct and meet customer expectations.” (Vargas, 2018, p. 44)*

Electronic board inspection is directly related to product quality. Company X therefore monitors process defects and machine inspection performance using internal quality indicators. Key performance indicators (KPIs) connect measurements with objectives and provide information for decision-making.

> *“A metric is the measure of something, that is, a measurement system that quantifies an attribute or characteristic. To create a key performance indicator, we need to associate an objective or goal that we intend to achieve, that is, a metric + an objective = key performance indicator or KPI. A metric, when compared to an objective, a reference or a baseline, becomes an indicator, which will show whether the goal was achieved or not.” (Almeida and Neto, 2015, p. 38)*

Indicators help identify critical issues and establish a reference for equipment evaluation. In this study, Company X sets a target of 98% approval in inspection tests, allowing 2% false failures associated with manufacturing tolerances and variation. This is an evaluation target, not a reported achieved result or a demonstrated defect-detection rate.

Once the process has been mapped and its problems, data and indicators examined, the evaluation criteria can be defined. The breadth of criteria reflects the need to address the existing inspection problems without introducing new operating difficulties.

The main requirements and their intended purposes are summarised below.

Requirement	Purpose
Test coverage, camera resolution, side cameras and component inspection height	Inspect the required component types and board layouts, addressing the existing inspection limitations.
False failures, shared library and repeatability	Meet the internal target of 98% approval in tests.
Programming time and technical support expertise	Provide practical, rapid programming and experienced support for a wide product range.
Inspection time	Match the assembly-line cycle time of 20 seconds or less.
Ease of operation	Support operator understanding and adaptation.
Machine size and weight	Fit the line layout and factory transport requirements.
Minimum and maximum board dimensions and thickness	Accommodate the range of electronic boards to be inspected.
Installed machine base, calibration and warranty	Support the evaluation of after-sales service.

Source: the author. Requirements translated from the original article.

2.3 Testing and comparing equipment

In addition to defining the criteria, the evaluator must consider the importance of each requirement. This requires knowledge of both the machines and the process in which they will operate. In the study described, Company X assigned a team to test each candidate machine for two weeks. Practical testing allowed the team to assess equipment against the process characteristics and requirements.

A comparison matrix can support this stage by bringing the assessment criteria and scores together. The Siqueira Campos bulletin describes the Pugh matrix as follows:

> *“Thus, this matrix encourages the comparison of several different concepts with a base concept, creating stronger concepts and eliminating weaker ones until an optimal concept is achieved. The matrix is widely used in projects, but its use has intensified in other fields, for example, to compare alternative solutions for any implementation.” (Boletim da Siqueira Campos Causa e Efeito, 2010, no. 26)*

Table 1 reproduces the article’s illustrative comparison of machines A, B and C. All scores are fictitious and demonstrate the method; they are not supplier test results. The original table uses scores of −1, 0 and 1 and equal displayed weights of 5.55% per criterion. The comparison basis or requirement is listed for each row.

Table 1 Illustrative machine comparison matrix

Criterion	A	B	C	Comparison basis
Test coverage	1	1	-1	1 good; 0 medium; −1 poor
Camera resolution	1	1	1	<20 microns
Side cameras	1	1	0	Available
Component height limit	1	1	1	20 mm
False failures	1	0	1	<2%
Shared library	1	0	1	Available
Repeatability	1	0	1	1 good; 0 medium; −1 poor
Programming time	1	0	1	2 hours
Technical support	1	-1	1	1 good; 0 medium; −1 poor
Inspection time	1	1	1	<20 seconds
Ease of operation	1	1	1	1 good; 0 medium; −1 poor
Machine dimensions	0	0	1	Smaller width
Machine weight	1	1	1	<2 tonnes
Board thickness range	1	1	1	0.8–4 mm
Board size range	1	1	1	500 × 460 (unit not stated)
Installed machine base	1	0	1	>5 machines
Calibration	1	1	0	Annual
Warranty	1	1	1	3 years

Source: the author. All values are illustrative. Each criterion has the original displayed weight of 5.55%.

Interpreting the illustrative comparison

The weighted total is the sum of each score multiplied by 0.0555. The original totals are retained below. In this example, machine A ranks first, followed by C and then B.

Machine	Original weighted total	Rounded to two decimals
A	0.9435	0.94
B	0.555	0.56
C	0.777	0.78

Numerical note: the 18 displayed weights of 5.55% sum to 99.90%, rather than exactly 100%. The totals above use those original weights and have not been recalculated using normalised weights. The original prose gave 0.55 for B and 0.77 for C; these have been corrected to 0.56 and 0.78 when rounded to two decimal places.

The requirements in the original matrix also use strict thresholds of <2% false failures and <20 seconds inspection time, whereas the narrative describes a 2% allowance and a cycle time of 20 seconds or less. Both formulations are retained here; an operational test specification would need to state explicitly whether the boundary values are acceptable.

3 Final considerations

The literature review and the equipment-evaluation study highlight the importance of understanding the manufacturing process before selecting machinery. Process mapping, problem analysis, performance requirements and practical testing provide a structured basis for comparing alternatives. These tools support decisions that reflect the needs and constraints of the production environment.

The article begins with a shortlist of machines already selected for evaluation. Further research could examine how to develop that shortlist from the wider supplier market. It could also investigate how to identify and organise the necessary project stages for different applications, priorities and time constraints.

The article presents an evaluation approach and an illustrative comparison. It does not report measured post-implementation improvements or a financial return on investment.

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